Presented by : OFF-GRID · SHAILESH KUMAR GUPTA

Sketch to 3D.

Hackathon Pitch
2D sketch to textured
3D asset

Sketch to 3D

Problem Statement

THE BARRIER

Meshes, texturing and rendering need specialist skills and software.

THE GAP

A sketch has no depth, hidden sides or surface detail. Something must infer them.

OUR GOAL

Sketch in, prompt optional, real .glb out, with the shape still recognisable.

Sketch to 3D

Our Pipeline

01 Clean

OpenCV: denoise, threshold, crop, centre. Thick strokes kept solid.

02 Refine

ControlNet scribble turns sketch plus prompt into a product-style image.

03 Reconstruct

TRELLIS: background removal, geometry and texture, exported as .glb.

04 Validate

trimesh: non-empty, real volume, textured. Then preview and download.

Light backend, free GPUs

Browser
HTML canvas and model-viewer: draw, upload, prompt, 3D preview.

Backend
Flask on Render (CPU): preprocessing, job queue, validation.

GPU
Hugging Face Spaces: ControlNet and TRELLIS inference.

Disclosed models: Stable Diffusion 1.5 + ControlNet, Microsoft TRELLIS. No weights stored locally.

What we measured

A real .glb with geometry and texture, checked by trimesh before download. The time covers cleaning, image generation, 3D reconstruction and validation.

57 s

Sketch to downloadable .glb (chair, warm run)

[N]

faces in the mesh

Sketch to 3D

Built for a live test

01 Fallbacks

Tries several TRELLIS Spaces in order. Without ControlNet it uses a filled sketch.

02 Fidelity

An edge-match check rejects generated images that drift away from the drawing.

03 Limits

Hidden sides are inferred. Free GPU quota is shared. Bold, closed outlines work best.

Thank You

Live demo: draw, generate, rotate, download .glb

- Live app: shailesh-kumar-gupta-06.github.io/sketch-to-3d
- Code: github.com/shailesh-kumar-gupta-06/sketch-to-3d
- Contact: real.shailesh.kumar.gupta@gmail.com